



BIRLA INSTITUTE OF TECHNOLOGY AND SCIENCE, Pilani
Pilani Campus
AUGS/ AGSR Division

FIRST SEMESTER 2026-27

COURSE HANDOUT

Date: 01/08/2026

In addition to part I (General Handout for all courses appended to the Time table) this portion gives further specific details regarding the course.

Course No : PHY F312
Course Title : STATISTICAL MECHANICS
Instructor-in-Charge : Tapomoy Guha Sarkar

Instructor: Tapomoy Guha Sarkar

1. Course Description:

This is a first course on Statistical mechanics aimed at students who have familiarity with classical mechanics, quantum mechanics, and classical thermodynamics. The stress is on developing the essential techniques of equilibrium statistical mechanics for non-interacting systems.

Classical Thermodynamics, as a study of macroscopic properties of materials, is no doubt an incredibly elegant interface between us and nature. Historically born out of the remarkable power of empiricism in the nineteenth century, the laws of Thermodynamics have stood the test of time. In its breadth of generality and profundity, the subject has found applications in almost all areas of fundamental and applied sciences. Here, we must note that scientific pedagogy is not necessarily chronologically ordered. The fast development in our understanding of matter at the microscopic level through advances in the atomic theories in the twentieth century required classical thermodynamics to find reconciliation with physics at the microscopic level. The role of statistics in bridging the laws of mechanics with those of thermodynamics is probably as profound as the laws themselves. In a first course on statistical mechanics aimed towards a target audience comprising undergraduate students of basic sciences and engineering it is important to reveal the subject in its broad, multi-layered beauty. The course aims to look at Thermodynamics as an emergent phenomenon and emphasize the role of statistics in bridging the microscopic and macroscopic descriptions.

2. Scope and Objective of the Course:

- ❖ Develop familiarity with the basic techniques of equilibrium statistical mechanics.
- ❖ Understand the role of probability theory and statistics in relating microscopic and macroscopic descriptions of thermodynamic systems
- ❖ Learn key methods of ensemble theory
- ❖ Distinguish between quantum and classical statistical mechanics and compute essential properties of ideal Bosonic and Fermionic systems.

3. Text Books:

TB1: Statistical Mechanics, Second Edition, R. K. Pathria, Paul Beale, 4th Edition, Academic Press



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4. Reference Books :

- R1. Thermodynamics and Statistical Mechanics, Greiner, Neise, and Stocker, Springer (1995)
- R2. Statistical Mechanics, Second Edition, Kerson Huang, WSE, Wiley (2000)
- R3. Fundamentals of Statistical and Thermal Physics, First Edition, F. Reif, McGraw-Hill (1965)
- R4. Thermal Physics by Charles Kittel and Herbert Kroemer, W. H. Freeman; Second edition (1980)

5. Course Plan (L : Lecture, T : Tutorial)

Module Number	Lecture session/Tutorial Session.	Reference	Learning Outcome
1. From Mechanics of particles to Statistical description. Emergent phenomenon (Number of lectures: 1)	L 1.1 The need for statistical Mechanics. Phase space. CM, QM description, and need for statistical description. Random walk problem as an emergent phenomenon. Introduction to how complex problems can be tackled using stochastic modeling Can probability theory give emergent laws?	R3: Chapter 1	The students shall be able to identify the role of emergent phenomena in probabilistic systems using the random walk problem. They shall be able to analyze a whole set of apparently dissimilar complex emergent phenomena using the probabilistic/statistical model.
	T1. Examples of random walk, Bernoulli trials, and the Binomial distribution. Brownian motion and Langevin equation.	Worksheet	
2. Thermodynamics and Kinetic Description.(Number of lectures: 7)	L 2.1 Macroscopic and Microscopic descriptions. Equilibrium Thermodynamics. Quasistatic processes. Adiabatic and diathermal walls (definitions) The concept of temperature from the Zeroth Law of	TB: 1.1, 1.2 R2 Chapter 1, 2, 3 R3	Students shall be able to use the laws of thermodynamics for simple hydrostatic and magnetic systems.





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	thermodynamics. Empirical temperature scales. Dilute gases and ideal gas temperature. Thermometry. Equation of state. Internal Energy. How to understand the concept of temperature in classical thermodynamics?	Chapter 3	- Students shall be able to interpret thermodynamic quantities as averages. - The concept of the distribution function shall be assimilated, and the students shall be able to compute statistical averages using the distribution function. - The students shall be able to use basic results from the non-interacting kinetic theory of dilute gases in diverse situations and appreciate the relationship between the microscopic and macroscopic descriptions.
	L 2.2 Introduction to heat, work, and internal energy. First law of thermodynamics. Applications of the first law of Thermodynamics. If energy conservation is known in mechanics, what is the need for the first law of thermodynamics? What is heat?	R3 Chapter 4	
	L2.3 Kinetic theory of dilute gases. Pressure, temperature, and internal energy.	Class notes	
	L2.4 Description of dynamics in Phase space, Mu-space, and distribution function. Maxwell's velocity distribution	R2: Chapter 6	
	L 2.5 Carnot's engine, The Second Law of Thermodynamics, Entropy, and Clausius's inequality. What is the limitation of the first law?	R2 Chapter 2	





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	What is entropy?		
	T 2.1 Problems on thermometry, Applications of the first law of Thermodynamics	R3 Chapter 4.1	
	L 2.6 Thermodynamic potentials and mathematical formulation of 2 nd Law. Applications to simple systems. Exact and inexact differentials, Maxwell's relations, and response functions	R2 Chapter 2, 3	
	L 2.7 Summary of thermodynamics, including the Third law and applications to ideal gas. Micro state and micro state	Class notes	
	T2.2 Problems involving the second law of Thermodynamics, heat engines, and thermodynamic potentials	Worksheet	
3. Ensemble Theory I Microcanonical ensemble MCE (Number of lectures: 5)	L3.1 Thermodynamic state of a system. Microstate and macrostate. Counting, probability, and statistics	R2 Chapter 6 R3: Chapter 2	- The students shall be well versed with the essential tools of counting, probability, and statistics (revision) - The students shall be able to count states and compute Boltzmann entropy
	L 3.2 The Gamma space and phase space distribution function. Ensembles. Liouville's theorem, Ergodicity and averages. Postulate of equal <i>a priori</i> probability. Microcanonical ensemble (MCE).	TB: 2.1, 2.2	
	T3. 3 Counting of states in MCE. Entropy and second law of	TB 2.3, 2.4	



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	thermodynamics (revisit) How do we understand entropy? How do we understand temperature? Can one derive the equation of state of an ideal gas from the first principles?		
	L 3.4 Sackur Tetrode equation and calculation of entropy using MCE. Gibbs' paradox and entropy of mixing.	R3 Chapter 7.3	
	L 3.5 The Ising model and calculation of thermodynamic quantities.	Class notes	
	T 3.1 Problems on counting and Stirling approximation. Entropy calculation using MCE.	R3 Chapter 4.5	
4. Ensemble theory II Canonical ensemble CE (Number of lectures: 10)	Is there a general method to derive mean macroscopic properties and their fluctuations using probability modeling? L4.1 Need for canonical ensemble and derivation of the distribution function for canonical ensemble	TB: 3.1, 3.2	- The students shall be able to calculate the canonical partition function for simple systems and compute various thermodynamic quantities. - The students shall be able to use the law of equipartition
	L4.2. The Canonical partition function. Calculations of the partition function for simple systems.	R3: Chapter 6 TB 3.3, 3.4, 3.5 R2	





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		Chapter 12	of energy to solve problems The students shall be exposed to magnetic systems
	L4.3 Method of most probable values.	TB 3.1, 3.2	
	L 4.4 Fluctuations in statistical mechanics.	R2 12.5	
	L4.5 Calculation of thermodynamic quantities using Canonical ensemble	TB Chapter 3	
	L4.6 The principle of Equipartition of energy	TB 3.7	
	L4.7 The Virial expansion	TB 3.7	
	L4.8 The problem of a collection of SHOs.	TB 3.8	
	L4.9 The theory of Paramagnetism	TB 3.9 R3: Chapter 7.8	
	L4.10 Paramagnetism (continued)	R3: Chapter 7.8	
	T4. 1 Problems on canonical ensemble. Calculation of partition function and thermodynamic quantities in hydrostatic and magnetic systems.	Worksheet	
5. Ensemble Theory III Grand Canonical Ensemble GCE	L5.1. Grand canonical distribution and Grand canonical partition function. The role of the Chemical potential.	TB 4.1, 4.2, 4.3 R3:	The students shall be able to compute the grand canonical partition function and obtain quantities of





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(Number of lectures: 3)		Chapter 6.9	interest in thermodynamics.
	L 5.2 Calculation of thermodynamic quantities using GCE and other variants of the ensemble.	R2 Chapter 13	
	L 5.3 Applications of GCE	TB Chapter 4	
	T5.1 Calculation of grand canonical partition function and obtaining thermodynamic quantities from it.	Worksheet	
6. Quantum Statistical Mechanics (Number of lectures: 3)	How does quantum mechanics change our view of statistical mechanics? L6.1. Distinguishable vs indistinguishable particles. Introduction to Quantum Statistical Mechanics. Revisiting the ensembles. Introduction to Bosons and Fermions. Spin-statistics theorem (no proof)	R3 : Chapter 9 R2: Chapter 8	- The students shall be able to distinguish between Bosonic and Fermionic particles, and also obtain the classical limit - They shall be able to grasp the idea of thermal wavelength
	L6.2. BE and FD distribution function using the method of the most probable distribution. The theory of Lagrange multipliers will be discussed in detail.	R3: Chapter 9.3	
	L 6.3 Classical limit of Quantum statistical mechanics and validity of Boltzmann's theory. The density matrix formalism for the description of Quantum systems.	R3: Chapter 9.8	
	What are Bose particles and		





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	Fermi particles?		
	T6. Problems with the FD and BE statistics	Worksheet	
7. BOSE systems (Number of lectures: 6)	L7.1. The ideal Bose gas.	TB 7.1 R2 Chapter 10	- The students shall be able to calculate properties of non-interacting Bosonic systems - The students shall be able to obtain the Plank's law from first principles and shall be able to apply it in various problems involving blackbody radiation
	L7.2. Bose-Einstein Condensation	R2: Chapter 11	
	L 7.3 Theory of Photon statistics and black body radiation.	TB 7.2 R3 Chapter 9.13	
	L 7.4 Photon statistics and connection with quantum optics description.	R3 Chapter 9.14, 9.15	
	T 7.1 Problems on blackbody radiation.	Worksheet	
	L 7.5 Lattice vibrations and Phonons.	R3: Chapter 10.1	
	L 7.6 Specific heat of solids (Debye and Einstein's theory)	R3: Chapter 10.2	
8. FERMI systems (Number of lectures: 3)	L 8.1 The theory of the Fermi gas	TB 8.1 R2 Chapter 9	The students shall be able to compute properties of non-interacting electrons
	L8.2. Degenerate electron gas in	R3	





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	metals	Chapter 9.16, 9.17	Low temperature linear dependence of electronic specific heat
	L8.3 Degeneracy pressure and Astrophysical application in stellar astrophysics	Class notes	
	T 8.1 Problems on Fermi gas theory	Worksheet	
9. Phase Transitions (Number of lectures: 2)	L 9.1 Thermodynamic description of phase transitions. The concept of order parameter. First-order and second-order transitions. Clausius-Clapeyron equation	R2 Chapter 4	The students shall get an exposure to the basic physics of first-order and second-order phase transitions
	L9.2 Second-order phase transitions and critical phenomenon (elementary) Scaling of response functions.	R2 Chapter 4	
	T 9.1 Problems on phase transitions	Worksheet	

TOTAL NUMBER OF MODULES: 9 . TOTAL NUMBER OF LECTURES : 40

6. Evaluation Scheme:

Component	Duration	Marks/Weightage (%)	Date & Time	Nature of component (Close Book/ Open Book)
Mid-Semester Examination	90 mins.	60 /30	TBA	Open/ Closed
Comprehensive Examination	180 mins	80/ 40	TBA	Open/ Closed
Class Tests/Assignments	40min. $\times$ 4 best three to be considered	60/30	TBA	Open/ Closed

7. Online Consultation Hour: To be announced in the class.

8. Notices: Will be displayed only on **Google Classroom/Nalanda** website.

9. Make-up Policy: As per Institute rules



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10. Note (if any): NA

11. It is the responsibility of the students to keep all the evaluated and returned answer sheets till the end of the semester.

Instructor-in-charge
Course No. PHY F111



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